

CHUR, Switzerland

WMO: 067860

Lat: 46.8703N

Lon: 9.5306E

Elev: 1824

StdP: 13.75

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.0	20.1	7.4	8.6	20.8	11.2	10.4	23.2	20.5	40.4	18.3	40.9	3.8	200	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	88.0	66.6	84.3	65.4	80.8	64.5	68.3	83.5	66.8	80.8	65.5	78.1	10.5	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.3	93.2	72.2	62.0	89.0	71.2	60.9	85.3	70.6	33.5	83.5	32.3	81.4	31.3	78.2	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	16.2	14.5	DB	11.8	93.2	4.8	3.1	8.3	95.4	5.5	97.3	2.7	99.0	-0.8	101.3	(4)
(5)			WB	10.4	70.8	4.7	1.1	7.0	71.6	4.3	72.2	1.6	72.8	-1.8	73.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	34.6	37.1	44.4	51.2	58.6	64.8	67.6	66.9	59.3	52.9	43.0	36.2		
	DBStd	13.60	7.15	7.98	7.39	7.09	6.67	7.12	6.26	6.13	6.22	6.81	7.65	6.99		
	HDD50	1841	479	363	201	73	9	1	0	0	5	48	234	428		
	HDD65	5299	943	782	640	416	217	91	45	50	188	375	660	892		
	CDD50	2377	1	2	26	107	276	446	545	523	285	140	24	2		
	CDD65	359	0	0	0	1	19	86	125	108	18	2	0	0		
	CDH74	2963	0	0	0	20	202	769	1030	820	112	10	0	0		
	CDH80	908	0	0	0	0	38	239	346	273	12	0	0	0		
(14) Wind	WSAvg	5.9	5.2	5.8	6.5	7.2	6.5	6.3	6.1	5.5	5.2	5.2	5.5	5.3		
(15) Precipitation	PrecAvg	43.1	2.5	2.6	2.7	2.7	3.7	4.8	5.1	5.6	3.8	3.1	3.5	2.9		
	PrecMax	59.7	6.6	9.0	7.0	5.9	6.4	9.3	8.5	9.4	7.4	7.2	11.4	7.8		
	PrecMin	32.5	0.2	0.6	0.5	0.6	1.1	1.9	2.3	0.9	1.2	0.0	0.0	0.2		
	PrecStd	6.2	1.7	2.0	1.5	1.3	1.6	1.9	1.5	1.8	1.7	1.7	2.2	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	60.3	68.9	76.8	84.3	90.8	92.3	92.4	81.1	74.9	65.8	57.4		
		MCWB	44.0	45.4	51.0	56.1	61.5	66.8	67.3	67.1	63.6	58.0	52.1	46.3		
	2%	DB	50.8	55.4	64.3	72.3	79.4	86.3	87.9	86.8	77.3	70.3	61.3	52.6		
		MCWB	41.9	43.7	48.9	54.1	60.0	65.6	66.8	66.9	62.9	56.6	49.8	43.8		
	5%	DB	47.2	51.7	60.3	68.4	75.4	82.3	83.9	82.4	73.6	66.9	57.5	48.8		
		MCWB	40.1	41.5	47.1	52.4	58.2	64.5	65.9	66.4	61.1	55.3	48.2	41.8		
	10%	DB	43.9	47.7	56.4	64.3	71.3	78.3	79.8	78.2	70.1	63.3	53.9	45.5		
		MCWB	38.3	39.9	45.5	50.3	56.8	63.3	64.8	64.8	59.5	53.9	46.6	39.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	46.7	52.2	57.5	64.2	69.4	70.2	70.2	66.2	60.4	53.1	47.3		
		MCDB	52.5	57.2	65.2	74.2	80.2	86.0	87.4	85.3	77.5	69.3	62.6	55.1		
	2%	WB	43.0	44.3	49.9	55.1	61.6	67.2	68.1	68.2	64.0	58.3	50.7	44.9		
		MCDB	49.0	53.5	61.9	70.1	75.8	82.4	83.7	82.7	74.4	66.8	59.0	51.1		
	5%	WB	40.8	42.3	48.0	53.1	59.4	65.5	66.7	66.7	62.1	56.6	49.0	42.8		
		MCDB	45.9	50.1	58.4	65.9	71.8	79.6	80.9	79.5	71.6	64.3	56.2	47.5		
	10%	WB	39.0	40.3	46.2	51.2	57.6	63.8	65.3	65.2	60.4	54.9	47.2	40.8		
		MCDB	43.1	46.8	54.6	62.6	68.9	76.3	77.9	76.5	68.0	62.0	53.3	44.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	13.9	17.0	18.5	18.4	18.6	18.6	17.5	16.3	15.9	12.7	11.1		
		MCDBR	15.3	19.8	23.7	25.3	25.2	25.3	25.2	24.2	21.3	20.7	17.6	15.4		
	5% WB	MCWBR	9.2	11.0	12.2	11.8	10.7	10.3	9.5	9.9	10.4	10.3	9.2	8.9		
		MCWBR	13.3	18.2	21.7	23.3	22.1	23.0	22.3	21.7	19.4	18.0	16.0	13.2		
(40) Clear-Sky Solar Irradiance	taub		0.266	0.293	0.339	0.388	0.407	0.416	0.415	0.406	0.371	0.339	0.300	0.265		
	taud		2.331	2.262	2.215	2.205	2.250	2.279	2.284	2.325	2.401	2.464	2.464	2.384		
	Ebn at Noon		264	277	279	274	272	270	268	266	266	259	250	251		
	Edh at Noon		23	31	38	42	42	41	40	37	31	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		457	750	1134	1401	1540	1793	1770	1540	1183	772	461	379		
	RadStd		55	92	159	217	155	162	157	145	118	89	65	52		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.68	N/A	N/A	+2.09	+0.85	+0.65	N/A	N/A	+157	+83
(47) Regional (116 neighbors)		N/A	N/A	N/A	+1.22	+0.72	+0.59	N/A	N/A	+106	+29

Nomenclature: See separate page